

SQL PROJECT

PG details which includes rooms and their types, residents residing in the room, food menu and their ratings, maintenance works and their status, rent payment and details, attendance for residents, complaints raised by residents.

Database creation:

```
1 • create database PG;
2 • use PG;
```

Output			
Action Output			
#	Time	Action	Message
✓ 1	21:09:44	create database PG	1 row(s) affected
✓ 2	21:11:31	use PG	0 row(s) affected

Table creation:

```
3 • create table residents(resident_id int Primary key Not null auto_increment, name varchar(100) not null, age int not null, gender varchar(50) not null,
4 phone int not null, email varchar(100) not null, joining_date date not null, room_id varchar(50) not null, status varchar(50) not null);
5 • create table rooms(room_id int primary key not null auto_increment, room_number int not null, floor int not null, room_type varchar(50) not null,
6 capacity int not null, monthly_rent int not null, status varchar(50));
7 • create table food_menu(meal_id int primary key not null auto_increment, meal_date date not null, meal_type varchar(100), dish_name varchar(250) not null,
8 category varchar(100) not null);
9 • create table food_ratings(rating_id int not null primary key auto_increment, meal_id int not null, resident_id int not null ,
10 rating int not null, feedback varchar(500), rating_date date not null);
11 • create table complaints(complaint_id int primary key not null auto_increment, resident_id int not null, room_id int not null,
12 category varchar(100), complaint_date date not null, status varchar(100), priority varchar(50) not null);
13 • create table maintenance(maintenance_id int primary key not null auto_increment, room_id int not null, category varchar(200),
14 description varchar(500), amount int not null, maintenance_date date not null, status varchar(50));
15 • create table payments(payment_id int primary key not null auto_increment, resident_id int not null, payment_type varchar(200),
16 amount int, payment_date date, payment_method varchar(50), payment_status varchar(100));
17 • create table attendance(attendance_id int primary key not null auto_increment, resident_id int not null, attendance_date date, status varchar(100));
18
```

✓ 3	21:15:38	create table residents(resident_id int Primary key Not null , name varchar(100) not null, age int not null, gender v...	0 row(s) affected
✓ 4	21:46:41	create table rooms(room_id int primary key not null, room_number int not null, floor int not null, room_type varchar...	0 row(s) affected
✓ 5	21:50:02	create table food_menu(meal_id int primary key not null, meal_date date not null, meal_type varchar(100), dish_n...	0 row(s) affected
✓ 6	21:52:01	create table food_ratings(rating_id int not null primary key, meal_id int not null, resident_id int not null, rating int n...	0 row(s) affected
✓ 8	22:02:04	create table complaints(complaint_id int primary key not null, resident_id int not null, room_id int not null, category ...	0 row(s) affected
✓ 9	22:04:37	create table maintenance(maintenance_id int primary key not null, room_id int not null, category varchar(200), de...	0 row(s) affected
✓ 10	22:07:23	create table payments(payment_id int primary key not null auto_increment, resident_id int not null, payment_type ...	0 row(s) affected
✓ 11	22:08:27	create table attendance(attendance_id int primary key not null auto_increment, resident_id int not null, attendanc...	0 row(s) affected

Alter Data:

```
18 • alter table residents modify room_id int not null;
```

✓ 13	22:19:29	alter table residents modify room_id int not null	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
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Adding Column:

```
alter table complaints add column description varchar(100);
```

16 22:23:47 alter table complaints add column description varchar(100)

0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

Inserting values:

```
insert into residents(name, age, gender, phone, email, joining_date, room_id, status) values
('Ananya', 24, 'Female', '9876543210', 'ananya@gmail.com', '2026-06-10', 1, 'Active'),
('Divya', 23, 'Female', '9876543211', 'divya@gmail.com', '2026-06-15', 2, 'Active'),
('Harini', 25, 'Female', '9876543212', 'harini@gmail.com', '2026-07-01', 2, 'Active'),
('Keerthana', 24, 'Female', '9876543213', 'keerthana@gmail.com', '2026-07-05', 4, 'Active'),
('Meena', 26, 'Female', '9876543214', 'meena@gmail.com', '2026-07-10', 4, 'Active'),
('Nandhini', 23, 'Female', '9876543215', 'nandhini@gmail.com', '2026-07-15', 4, 'Active'),
('Pavithra', 25, 'Female', '9876543216', 'pavithra@gmail.com', '2026-08-01', 5, 'Active'),
('Ramya', 24, 'Female', '9876543217', 'ramya@gmail.com', '2026-08-05', 5, 'Active'),
('Sowmya', 23, 'Female', '9876543218', 'sowmya@gmail.com', '2026-08-10', 8, 'Active'),
('Swetha', 27, 'Female', '9876543219', 'swetha@gmail.com', '2026-08-15', 8, 'Active');
```

35 11:35:20 insert into residents(name, age, gender, phone, email, joining_date, room_id, status) values ('Ananya', 24, 'Fem...

10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

```
select * from residents;
```

[illegible]

```
select name, age, email from residents;
```

Result Grid		Filter Rows:	
	name	age	email
▶	Ananya	24	ananya@gmail.com
	Divya	23	divya@gmail.com
	Harini	25	harini@gmail.com
	Keerthana	24	keerthana@gmail.com
	Meena	26	meena@gmail.com
	Nandhini	23	nandhini@gmail.com
	Pavithra	25	pavithra@gmail.com
	Ramya	24	ramya@gmail.com
	Sowmya	23	sowmya@gmail.com
	Swetha	27	swetha@gmail.com

Where clause:

```
select * from residents where age > 24;
```

[illegible]

```
select * from residents where status = 'Active';
```

[illegible]

```
select * from complaints where status = 'Open';
```

	complaint_id	resident_id	room_id	category	complaint_date	status	priority	description
▶	2	2	2	Water	2026-09-20	Open	High	No water in bathroom
	5	5	4	Food	2026-09-21	Open	Medium	Dinner was too spicy
	7	7	5	Water	2026-09-22	Open	High	Water leakage
	9	9	8	Electricity	2026-09-23	Open	High	Light is not working
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Order By clause:

```
select name, age from residents order by age desc;
```

	name	age
▶	Swetha	27
	Meena	26
	Harini	25
	Pavithra	25
	Ananya	24
	Keerthana	24
	Ramya	24
	Divya	23
	Nandhini	23
	Sowmya	23

```
select dish_name, category from food_menu order by dish_name ASC;
```

	dish_name	category
▶	Chapati	North Indian
	Chapati	North Indian
	Curd Rice	South Indian
	Dosa	South Indian
	Idli	South Indian
	Lemon Rice	South Indian
	Pongal	South Indian
	Poori	South Indian
	Sambar Rice	South Indian
	Tomato Rice	Rice
	Veg Meals	South Indian
	Vegetable ...	Rice

```
select room_number, monthly_rent from rooms order by monthly_rent desc;
```

	room_number	monthly_rent
▶	101	12000
	301	12000
	102	9000
	103	9000
	202	9000
	302	9000
	201	7500
	203	7500

Updating Values:

```
update residents set status = 'Inactive' where resident_id = 10;
```

```
1 • select * from residents where resident_id = 10;
```

	resident_id	name	age	gender	phone	email	joining_date	room_id	status
▶	10	Swetha	27	Female	9876543219	swetha@gmail.com	2026-08-15	8	Inactive
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Delete clause:

```
delete from maintenance where maintenance_id = 8
```

63 11:54:56 delete from maintenance where maintenance_id = 8 1 row(s) affected

```
select * from maintenance;
```

64 11:55:42 select * from maintenance LIMIT 0, 1000 7 row(s) returned

	maintenance_id	room_id	category	description	amount	maintenance_date	status
▶	1	1	Plumbing	Bathroom tap repair	800	2026-09-18	Completed
	2	2	Electrical	Fan replacement	1800	2026-09-19	Completed
	3	4	Plumbing	Water leakage repair	2500	2026-09-20	Completed
	4	5	Electrical	Tube light replacement	600	2026-09-20	Completed
	5	8	Furniture	Chair repair	1200	2026-09-21	Completed
	6	4	Electrical	Switch board repair	900	2026-09-22	Pending
	7	2	Plumbing	Wash basin repair	1500	2026-09-23	Completed
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Count:

```
select COUNT(*) AS total_residents from residents;
```

65 11:57:22 select COUNT(*) AS total_residents from residents LIMIT 0, 1000 1 row(s) returned

	total_residents
▶	10

```
select COUNT(*) AS active_resident from residents where status = 'Active';
```

	active_resident
▶	9

Sum:

```
select SUM(amount) as total_maintenance_expense from maintenance;
```

67 12:00:24 select SUM(amount) as total_maintenance_expense from maintenance LIMIT 0, 1000 1 row(s) returned

	total_maintenance_expense
▶	9300

```
select SUM(amount) AS total_paid from payments where payment_status = 'Paid';
```

68 12:01:22 select SUM(amount) AS total_paid from payments where payment_status = 'Paid' LIMIT 0, 1000 1 row(s) returned

	total_paid
▶	82000

Average:

```
select AVG(rating) as average_food_rating from food_ratings;
```

69 12:02:27 select AVG(rating) as average_food_rating from food_ratings LIMIT 0, 1000 1 row(s) returned

	average_food_rating
▶	3.6154

```
select AVG(amount) as average_maintenance_cost from maintenance;
```

70 12:03:23 select AVG(amount) as average_maintenance_cost from maintenance LIMIT 0, 1000 1 row(s) returned

	average_maintenance_cost
▶	1328.5714

Aggregate functions with group by and where clause:

```
select room_id, COUNT(*) AS total_residents from residents group by room_id;
```

71 12:06:58 select room_id, COUNT(*) AS total_residents from residents group by room_id LIMIT 0, 1000 5 row(s) returned

	room_id	total_residents
▶	1	1
	2	2
	4	3
	5	2
	8	2

```
select meal_id, AVG(rating) AS average_rating from food_ratings group by meal_id;
```

72 12:08:07 select meal_id, AVG(rating) AS average_rating from food_ratings group by meal_id LIMIT 0, 1000 12 row(s) returned

	meal_id	average_rating
▶	1	4.3333
	2	2.3333
	3	4.5000
	4	3.5000
	5	4.5000
	6	2.5000
	7	4.5000
	8	2.5000
	9	4.5000
	10	2.5000
	11	4.5000
	12	3.5000

```
select meal_id, avg(rating) as average_rating from food_ratings group by meal_id having avg(rating) < 3;
```

✓ 73 12:10:40 select meal_id, avg(rating) as average_rating from food_ratings group by meal_id having avg(rating) < 3 LIMIT ... 4 row(s) returned

	meal_id	average_rating
▶	2	2.3333
	6	2.5000
	8	2.5000
	10	2.5000